The Spine

Parametric Design with AI. A ring-and-spine structure built from scratch in Grasshopper by prompting Raven, reconstructing the parametric logic of my own studio project.

ASSIGNMENT 03 · J. RUSSO

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June 7, 2026
Raven · Grasshopper

The Spine is a single NURBS curve carrying a row of circular rings that vary in size along its length, lifted on a cross-braced support structure. The geometry is reconstructed from my own studio project, and the objective was to rebuild its parametric logic entirely through natural-language prompts to Raven, an agentic text-to-Grasshopper assistant, rather than wiring components by hand. The definition was built in six labeled groups through five sequenced prompts: three detailed phased builds, one image-based refinement against the studio drawing, and one generative prompt that introduces degeneracy.

RAVEN PROMPT STRATEGY · THE PROMPTS ARE THE LOGIC

#	TYPE	PROMPT TO RAVEN	WHY THIS PROMPT, AND WHAT CHANGED
1	DETAILED	Brief: build a parametric "Spine", a row of differently sized rings threaded along a curve and lifted on a cross-braced support. Then: interpolate a NURBS curve through about six control points (Spine Length, Amplitude), and divide into N stations (Ring Count 12 to 60) with a perpendicular frame at each so rings are true cross sections, not flat circles.	Opens with the design intent, then sets the one rule the rest depends on, perpendicular cross sections, before any geometry. This is the brief.
2	DETAILED	Rings. Drive radius from the curve parameter through a Graph Mapper remapped between Min and Max Radius. Two concentric circles (Wall Thickness) extruded by Ring Depth into collars.	Names the size logic so Raven uses a Graph Mapper I can shape by hand, bounded Min to Max. Changed the skeleton into sized geometry and added four form sliders.
3	IMAGE-BASED	[studio drawing attached] Tune the Graph Mapper so ring sizes cluster into fat nodes matching the sheet, and adjust Max Radius and Ring Count. Tune only, do not rebuild.	The clustering rhythm is visual, faster shown than described, so I hand Raven my own drawing. Changed an even swell into the clustered node rhythm of the sheet.
4	DETAILED	Supports. Sample at Support Spacing. Where curve Z exceeds Support Height, drop a vertical member to Z 0 and cross-brace into a diagrid. Conditional on height.	Written as a conditional so structure appears only where the spine lifts off, not everywhere. Added the first height-driven constraint.
5	GENERATIVE	Seed. Jitter the radius profile and station positions within hard clamps (Min and Max radius, perpendicular frames, height threshold) to sample many structurally distinct, valid spines.	Stops prompting for one form and asks for a bounded space a seed can sample. Changed one fixed form into a family of valid solutions, degeneracy.

CORRECTIONS, THE DEBUGGING

Corrections issued when Raven drifted: flat rings to perpendicular frames, twist to Horizontal Frames, uniform radius to Remap Numbers wiring, supports everywhere to the height-threshold conditional.

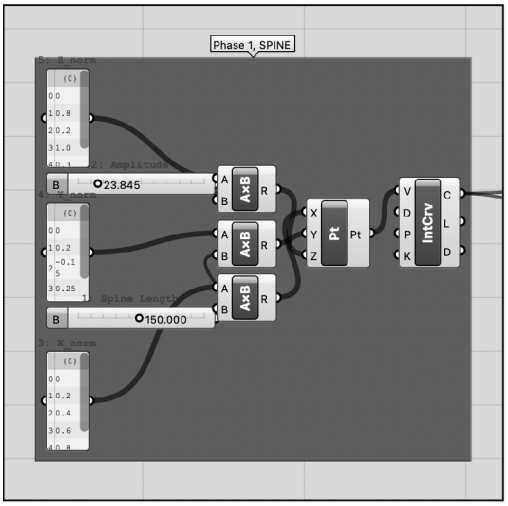
DEGENERACY, THE GENERATIVE MOVE

Degeneracy (Edelman and Gally) is the capacity of structurally different configurations to perform the same function. The final prompt asks Raven for a bounded solution space rather than a single form, which turns the definition from an object into a system.

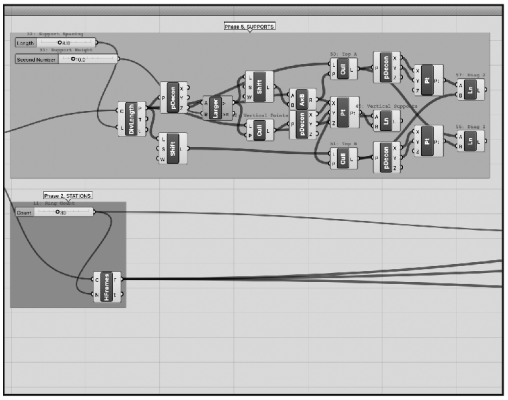
THE DEFINITION · SIX GROUPS, TEN SLIDERS, FOUR CONSTRAINTS

#	GROUP	FUNCTION	SLIDERS
1	**Spine**	Interpolated NURBS curve through control points	`Spine Length, Amplitude`
2	**Stations**	Even division with a perpendicular frame at each point	`Ring Count`
3	**Radius**	Graph Mapper drives radius, remapped to bounds	`Min Radius, Max Radius`
4	**Rings**	Two concentric circles per station, extruded to collars	`Wall Thickness, Ring Depth`
5	**Supports**	Cross-braced members drop to ground above a threshold	`Support Spacing, Support Height`
6	**Seed**	Jitters radius and node positions within bounds	`Seed`

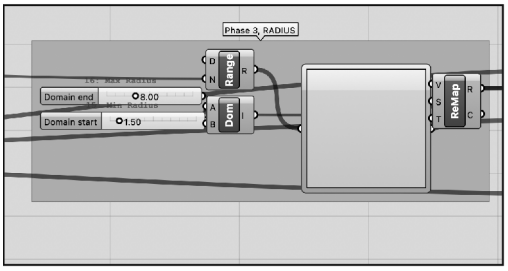
THE GRASSHOPPER CANVAS · GROUPED AND LABELED BY PHASE



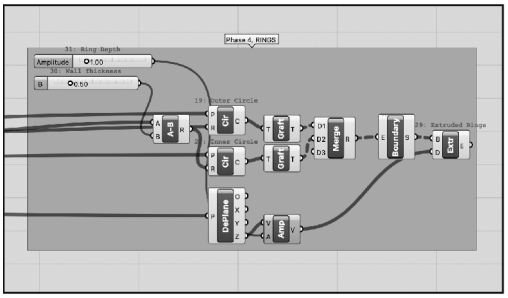
Phase 1, Spine. Number params to AxB to Pt to Interpolate Curve, with Spine Length and Amplitude.



Phases 2 and 5, Stations and Supports. Division, perpendicular frames and the cross-braced diagrid.



Phase 3, Radius. The Graph Mapper and Remap between Min and Max Radius.

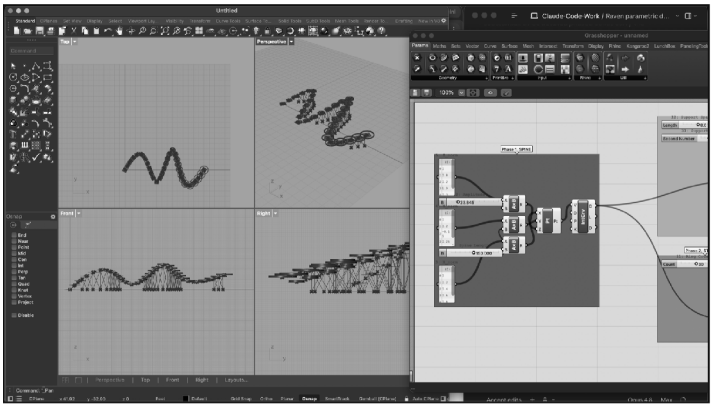


Phase 4, Rings. Concentric circles extruded into collars on each frame.

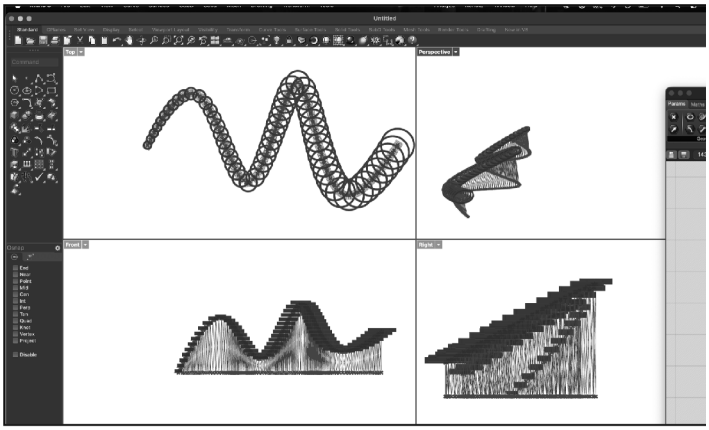
WORKING CONSTRAINTS · VERIFIED IN THE RUNNING DEFINITION

1. Rings are true perpendicular cross sections (Horizontal Frames, untwisted).
2. Radius is bounded between Min and Max, so the form never collapses or self-intersects.
3. Supports are conditional: a member appears only where curve Z exceeds the Support Height threshold.
4. The seed perturbs only within those bounds, so every sampled solution is valid by construction.

GENERATED GEOMETRY · TWO SLIDER VARIATIONS FROM ONE DEFINITION

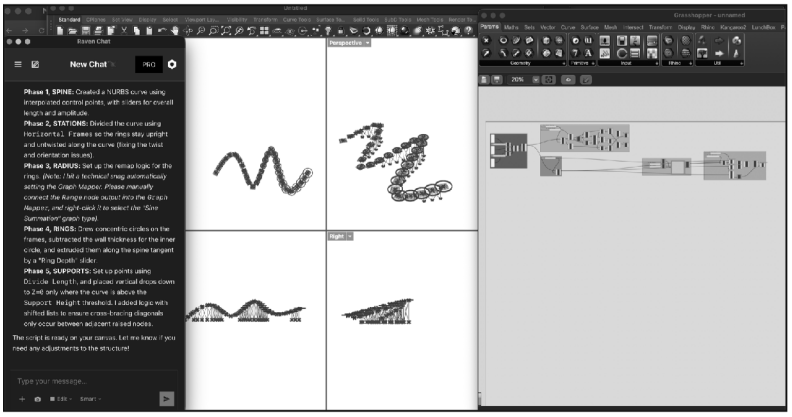


Variation 1. Lower Ring Count and a wider radius range, fewer larger rings.



Variation 2. Higher Ring Count and a tighter spine, many smaller rings.

THE RAVEN SESSION



Raven building the definition phase by phase, in Grasshopper.

WORKED. The five prompts built the spine, the stations, the rings and the supports, and the geometry matches the studio drawing. The Graph Mapper gave direct control of the size profile.

CORRECTED. Where Raven drifted, ring orientation and the support placement, the four corrections above brought it back.

SURPRISED. Raven selected components I did not name, a Graph Mapper for the radius profile and Horizontal Frames to keep the rings untwisted along the curve.

ONE CHANGE NEXT TIME. State the correctness constraints, perpendicular untwisted frames and supports only above the height threshold, inside the first detailed prompt rather than fixing them by correction. The constraint is the prompt.

DELIVERABLE, MAPPED TO THE RUBRIC

CRITERION	PTS	WHERE IT IS MET
Raven prompt strategy	3	Five prompts, the brief first, three distinct types (detailed, image-based, generative), rationale and what changed per prompt, four corrections. Page 1.
Grasshopper file	3	Six labeled groups, ten sliders, four working constraints, runs without errors. Submitted as the .gh.
Grasshopper documentation	2	Four labeled-group canvas shots (page 2), generated geometry and two slider variations (page 3).
Raven conversation summary	2	What worked, what needed correction, what surprised, a session screenshot and one change. Page 3.
Quality and effort	3	Own studio project as the seed, degeneracy as the generative thesis, the frames and twist insight.
Documentation format	1	Single consolidated PDF, three pages.
File naming	1	Valerie_Jimenez_Assignment03.pdf, .gh and .3dm, matching names.
Total	**15**	**Single consolidated PDF, technical files submitted separately.**

Submission set. Valerie_Jimenez_Assignment03.pdf · Valerie_Jimenez_Assignment03.gh · Valerie_Jimenez_Assignment03.3dm, matching names.